MISSION REPORT

Budapest I

Status:	ONGOING
Launch Date:	Year 2, Day 25
Operator:	ESA
Target:	Moho

This document contains technical specifications and flight data for the Budapest I mission as part of the Kerbin Expansion Initiative.

April 15, 2026

1. Mission objectives

The primary goal of the mission was confirming the theories about a Hohmann transfer to Moho in a reasonable time with the Δv budget of an Ariane 7 UH booster. Moho is well known for being hard to get to, especially because of its insertion burn. As such, Budapest I was tasked with finding out if it's possible to use the Ariane 7 UH to make a one-way trip to Moho from Kerbin in an optimal launch window. As a reminder, a launch window between Kerbin and Moho opens when the phase angle is about 110° (i.e. Moho is 110° ahead of Kerbin).

A secondary mission objective was testing the capabilities of the crew to transport and assemble a rover made for low-gravity in-situ. That is, the individual pieces of the rover would be stowed away in a storage container and assembled on-site by an engineer and subsequently used for excursions further away from the landing site if needed. Testing such a rover and certifying it would be very useful for future missions to Dres and Duna, even though they have higher gravity than Moho.

Although the primary goal is to test the one-way capabilities of the booster, a subsequent mission is to be organized to get the Kerbals back to Kerbin.

To summarize:

- Certify that the Ariane 7 UH booster has enough Δv for a one-way trip with a lander attached.
- Verify that the it is possible to travel and land to Moho in the established launch window with a Hohmann transfer.
- If the upper cryogenic stage has any fuel left after the insertion burn, keep it in orbit for later rendez-vous with the lander.
- Certify the rover can be built and that it works well in low-gravity situations.

More details about the rover can be found in the document detailing its performance, although a short version can also be found here.

2. Kerbonauts

Talk about the kerbonauts, if it is a manned mission. If it is not, skip this section.

3. Vehicle specifications

Describe the hardware for this mission. Here you can get the data for the specific craft used and describe any custom modifications for this mission, if any.

Table 1: Hardware Configuration: Vanguard-1 Lander

Component	**Mass (t)**	**Primary Engine**
Command Module	0.8	N/A
Descent Stage	4.2	LV-909 "Terrier"
Service Module	2.5	Poodle Mk2
Booster Stage	35.0	S1 SRB-KD25k

Phase	**Required Δv**	**Actual Δv**	**Efficiency**
Kerbin Ascent	3400 m/s	3550 m/s	95.7%
Munar Transfer	850 m/s	845 m/s	100.6%
Landing	600 m/s	720 m/s	83.3%

4. Δv budget

4. Post-mission analysis

Talk about the conclusion of the mission, describe what the failures were, what we can learn from this, this would also be the place to write a report about incidents or kerbal-error.

Appendix: Mission Imagery

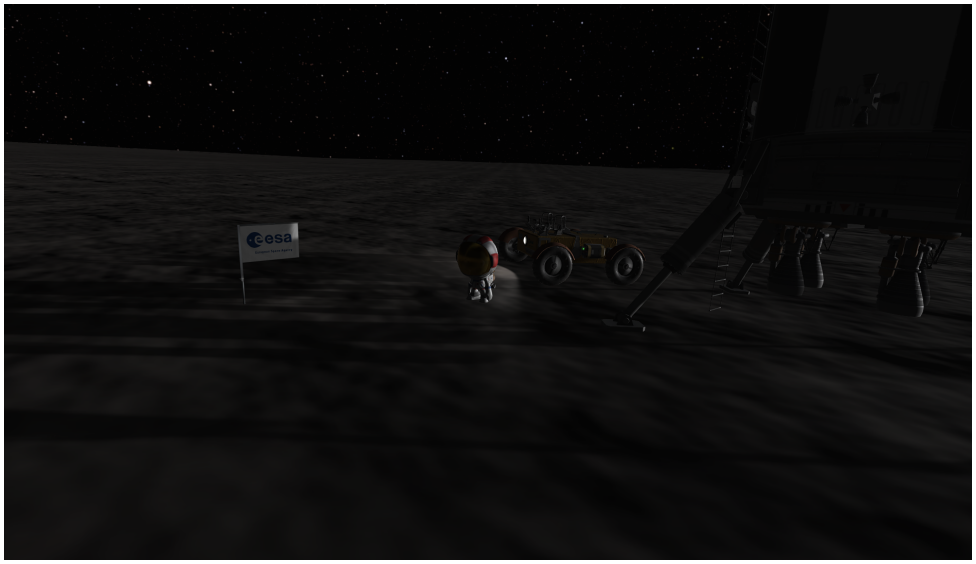

Figure 1: From left to right: the ESA flag, Dezer Kerman, the Columbus rover and the DEL lander in Dezer Crater, shortly after landing.